Calibrated date ranges for the Pentateuchal sources, and a direct measurement of what archaizing can buy: Biblical Hebrew and Ancient Greek

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Abstract

Linguistic dating of ancient texts assumes that a late author cannot convincingly write like an early one. Should that assumption fail, every argument for the antiquity of a biblical text fails with it. Both sides of the Classical/Late Biblical Hebrew debate have asserted an answer for decades, but neither has measured one. It is measured here, and then used to date the Pentateuchal sources. Features were extracted from 144,517 words of Biblical Hebrew anchored by synchronisms with Assyrian, Babylonian, and Persian records, partitioned into 281 passages of about 500 words so that high-frequency morphosyntax could be estimated within a passage rather than only across a whole book. Under leave-one-book-out validation, a weighted ridge model over 578 features gives a mean absolute error of 118 years against 137 for the constant predictor, a rank correlation of +0.66, and correct chronological ordering for 72.9% of 295 book pairs; a permutation null that re-runs the whole pipeline on shuffled dates gives $p = 0.003$. Nesting the configuration search inside the validation, so that nothing about the model is chosen using the book being scored, attenuates these to +0.62 and 71.5%. With distribution-free conformal intervals, the Priestly, Deuteronomic, and JE composites are placed after the exile with probability of at least 0.84, in the relative order JE $\rightarrow$ D $\rightarrow$ P, though none of them contributed to training. Archaizing was then priced two ways. Replacing every Late Biblical Hebrew form in six securely dated late books with its Classical counterpart moves their estimated dates by only +9 years, as no lexical item is worth more than 12.1 years per standard deviation. But the Second Sophistic Atticizers, who imitated Classical Attic under a formal curriculum of prose composition, are pushed 206 years too early by an equivalent model trained only on non-archaizing Greek. The price of archaizing depends on its mechanism: swapping vocabulary is cheap, while rebuilding a grammar is not. An archaizing account of an apparently early Hebrew text therefore requires a scribal training for which the Second Temple period offers no evidence.

Introduction

Scholars in biblical studies, classical philology, and Near Eastern linguistics routinely use internal linguistic evidence to constrain the dates of ancient texts. The transition from Classical Biblical Hebrew (CBH; also called Standard Biblical Hebrew) to Late Biblical Hebrew (LBH) is among the best-documented diachronic shifts in any ancient language [1–3], and comparable transitions are well attested in Ancient Greek [4] and Akkadian. Despite this rich empirical tradition, linguistic dating almost never produces calibrated probability estimates; scholars identify features as "archaic" or "late" and assign texts to qualitative categories, but the resulting claims carry no quantified uncertainty.

Beneath the practice sits an assumption that is stated often and tested never. The enterprise presupposes that a late author cannot convincingly write like an early one. Were deliberate archaizing easy, so that a Persian-period scribe could adopt Classical vocabulary and thereby acquire a Classical date, every linguistic argument for the antiquity of a biblical text would be worthless, and the objection would apply to traditional and computational methods alike. The dialectal school has pressed this point for two decades [5, 6], while the mainstream reply has been that sustained archaizing is difficult. Both positions are assertions. Neither side has put a number on the matter.

In this article I put a number on it, in two ways, and then use a model whose sensitivity to archaizing is known to date the Pentateuchal sources.

The dating model is built on 144,517 words of independently anchored Hebrew, partitioned into 281 passages of about 500 words. This partition matters more than it might seem. At the level of whole books, the corpus supplies 25 observations for hundreds of candidate features, a regime in which nothing can be estimated honestly. At the level of passages, the same corpus supplies 281 observations, and the features that survive are the high-frequency morphological and syntactic rates that can actually be measured in a few hundred words. Dates are reported as distribution-free conformal intervals calibrated on out-of-sample residuals rather than as point estimates.

Archaizing is then priced twice, by two routes that disagree informatively. The investigator can perform archaizing himself, so the ground truth is known exactly: securely dated *late* books have their LBH forms replaced by CBH counterparts at controlled rates, every feature is re-extracted from the altered text, and the result is dated by a model that never saw it. This prices the *lexical* mechanism, and the price is low. Ancient Greek then supplies what Hebrew cannot, namely texts whose authors really did archaize, deliberately and expertly, and whose dates of composition are securely known. The Second Sophistic Atticizers wrote imitation Classical Attic four to seven centuries after the fact, taught by a formal curriculum, well enough to be read as models of style for a millennium. This prices the *morphosyntactic* mechanism, and the price is an order of magnitude higher.

That contrast is the answer offered here to the standing objection. Archaizing is neither uniformly cheap nor uniformly hard; its cost depends on how deep into the grammar the imitation reaches, and the two depths differ by roughly a factor of twenty in their effect on an estimated date. Whether a given text could have been archaized thus turns on what its author was trained to do, which is a historical question about scribal education rather than a statistical one.

The results are bounded in both directions. The model orders books in time far better than chance and places the Pentateuchal sources after the exile with high probability, but its 68% intervals are 137 years wide on either side, and it cannot resolve dates within the Persian period. What it does resolve is whether a text falls before or after the exile, which is the question the Pentateuchal debate has actually turned on.

The Background section reviews the CBH/LBH distinction, the dialectal challenge, and the problem of circularity. The Data section describes the anchored corpus, the passage-level design, and feature extraction. The Model section presents the weighted ridge estimator, the variance-matching step, and the conformal intervals. The Validation section documents a failure mode that inflates reported accuracy in this literature and then reports leave-one-book-out performance against a permutation null. The Results section dates the undated units, and the section following it reports both archaizing measurements.

Background: Diachronic Biblical Hebrew

The CBH/LBH Distinction

Classical Biblical Hebrew (CBH) is the language of the pre-exilic prophetic corpus and early narrative material (8th–7th centuries BCE): Amos, Hosea, early Isaiah, and the narrative strands of Samuel–Kings. Late Biblical Hebrew (LBH) characterizes the post-exilic prose of Ezra, Nehemiah, Chronicles, Esther, and Daniel (5th–2nd centuries BCE).

The transition is marked by well-studied lexical, morphological, and syntactic changes [2, 3, 7, 8]. At the lexical level, the first-person singular pronoun shifts from *ankī* to *anī*; the relative pronoun *ašer* gives way to the enclitic *š*e-; the negator *lō'* is supplemented by and partly replaced by *'ēn*. At the morphological level, the wayyiqtol declines in frequency while the *hāyāh* + participle periphrastic increases. At the syntactic level, infinitive-absolute constructions rise, and sentence structure shifts toward analytic patterns associated with Aramaic influence [5, 6].

The consensus view [1, 9–11] holds that these features shift monotonically along the temporal axis, constituting a reliable if noisy diachronic signal. The present study adopts this view as a working assumption and then tests the principal alternative to it directly.

Diachronic or Dialectal

Young, Rezetko, and Ehrensvärd [5] argue that the CBH/LBH contrast may reflect geographic dialect, scribal tradition, or literary register rather than chronology. The Qumran sectarian texts (∼150 BCE) deliberately employ CBH vocabulary and morphology while simultaneously using Aramaic loan-words — a pattern they interpret as literary register rather than temporal reality. Hurvitz and the diachronic mainstream [7] counter that the dialectal alternatives require additional auxiliary hypotheses, and that the Qumran archaism case presupposes the original CBH features had already receded, confirming rather than undermining the diachronic interpretation.

This debate has been conducted almost entirely through argument. Its empirical core, however, is a quantitative question: how much apparent antiquity does deliberate archaizing actually purchase? That question is answerable. A model that assigns dates can be fed texts whose archaizing status is known, either because the investigator performed the archaizing or because the author's imitative program is a matter of literary-historical record. Both experiments are reported below. Neither resolves the dispute, but each supplies a measurement that either side can now argue from, which is more than the debate has had so far.

The Problem of Circularity

Dating methods that use features identified by comparing texts of assumed date risk circularity: the assumed dates shape the feature set, and the feature set is then used to infer unknown dates. Three safeguards are adopted.

A distinction is necessary before the safeguards can be stated, because the objection applies with very different force to different kinds of evidence. Dating a text by its resemblance to other Hebrew of supposed date is circular for the present purpose, since it invokes the very cline the model is meant to estimate. Dating a text by a datable referent is not. Esther's Achaemenid court and administration, the fall of Nineveh behind Nahum, and the Persian administrative vocabulary of Ecclesiastes are all anchored to events and language contacts documented outside the biblical tradition. That Esther cannot be pre-exilic follows from Persian history, not from Hebrew style, and the inference would stand were the language of Esther entirely unremarkable. The same logic underwrites Ezekiel's regnal dates. They are evidence because the kings and years can be checked against Babylonian records; had he invented them, they would be worth nothing.

The circularity that has damaged linguistic dating runs in the other direction, and it runs through the Torah. The pre-exilic standard against which lateness is measured was itself constructed in part from Torah texts assumed to be pre-exilic, and those texts were then dated by their conformity to it [5, 6]. Nothing in the present design enters that loop. The model learns the feature-to-date mapping from the anchored corpus alone, no Torah text is used in training, and every Torah estimate is therefore a genuine out-of-sample prediction.

With that distinction in place, three safeguards apply.

First, every training date rests on a synchronism or on a datable historical referent. Twelve of the 25 units carry an explicit synchronism, a named ruler or regnal year checkable against Assyrian, Babylonian, or Persian records; twelve more are fixed by a datable event, institution, or language contact. One unit, Jonah, is dated principally on typological grounds, and its removal is reported in Table 2.

Second, no feature is selected using date information. The feature set is fixed in advance by frequency rank in the corpus as a whole (Data section), not by correlation with date. Feature-by-date screening was evaluated and rejected: at this dimensionality the false discovery proportion among screened features is high enough that screening manufactures apparent signal (S2 Table).

Third, all reported performance comes from leave-one-*book*-out cross-validation in which standardization, weighting, regularization-parameter selection and interval calibration are performed inside each fold, and significance is established by a permutation null that re-runs the entire pipeline on shuffled dates.

Why an anchored corpus. The Hebrew prophetic books offer a form of historical anchoring available nowhere else in the Hebrew Bible: their superscriptions and internal chronological notices can be cross-checked against records entirely external to the biblical tradition. Ezekiel explicitly dates thirteen of his visions to regnal years of the Babylonian exile — "the fifth year of King Jehoiachin's captivity" (Ezek 1:2), "the sixth year" (8:1), and so forth — which correlate with Babylonian administrative texts independently dating Jehoiachin's deportation to 597 BCE. Other eighth- and seventh-century prophets (Amos, Hosea, Isaiah, Micah, Zephaniah, Nahum) are anchored by superscriptions naming Judahite and Israelite kings whose reigns are documented in the Assyrian and Neo-Babylonian annals. No comparable external anchoring exists for the Torah or the poetic books: their dates derive entirely from literary-critical argument, which is precisely the circularity this study seeks to avoid.

Asymmetric error and conservative upper bounds on antiquity. If the accepted dates for the anchored training texts were wrong, the error would be asymmetric: these texts can be *no older* than the historical events they reference, but they could in principle be *younger*. A training corpus misdated in the direction of being too early would calibrate the feature-to-date mapping at too early a point on the temporal axis, so unknown texts dated against that calibration would also be estimated as too early. Any systematic error in the training dates therefore propagates in a single direction: all model-derived dates shift toward greater recency. The dates reported here should be read as *conservative upper bounds on antiquity.* A text dated to 400 BCE may be more recent; the linguistic evidence provides no support for a date earlier than the estimate.

Prior Computational Work: Attribution versus Dating

Computational approaches to textual attribution and stylometric dating have a substantial history, from Mosteller and Wallace's Bayesian analysis of the Federalist Papers [12] through the stylometric tradition surveyed by Holmes [13] and the function-word methods of Burrows [14]. Among work applied specifically to the Hebrew Bible, Faigenbaum-Golovin et al. [15] provide the most methodologically rigorous prior computational treatment. Their Higher Criticism discrepancy method tests whether the distribution of lemma n-gram frequencies in a target text deviates from the null that it was produced by the same process as a reference corpus. Applied to 50 ground-truth chapters spanning D, the Deuteronomistic History, and P, it achieves 84% attribution accuracy in leave-one-out validation and yields interpretable discriminating lemma lists.

The present framework answers a different question, and three differences follow from that.

Continuous dates versus categorical assignment. The Higher Criticism method assigns each text to the most likely of three corpora. However, D and the Deuteronomistic History are both dated to the late monarchic or exilic period, so two texts composed at 650 and 590 BCE receive identical labels, and no intra-school temporal resolution is possible. The model used here returns a date with a calibrated interval, recovering whatever temporal resolution the data support and quantifying the rest as uncertainty.

Period features versus topic features. Faigenbaum-Golovin et al. operate on lemma and lemma n-gram frequencies, a representation in which content words — *gold* in Tabernacle chapters, *king* in regnal narratives — dominate. Content-word frequency is sensitive to topic and genre as much as to date. The feature set here is deliberately dominated by high-frequency grammatical rates (verbal aspect and stem distributions, part-of-speech transition rates, clause-relation profiles) that vary with period rather than subject matter. This is what permits comparison of poetry against narrative, which a bag-of-words representation cannot do.

A measured sensitivity to archaizing. Attribution methods offer no mechanism for telling a genuinely ancient text from one composed in a deliberately archaic register. A Persian-period author who archaizes to D norms will be assigned to D regardless of when he wrote. Rather than claim immunity from the problem, the sensitivity is measured directly below and reported as a quantity. A reader who thinks archaizing was widespread can take that exchange rate and work out how much of any reported date it might account for.

These are differences of purpose rather than deficiencies in either approach. The ideal treatment of a disputed text would apply both, as attribution output can set register priors while date estimates constrain the plausible settings for an attribution decision.

Data: Corpus and Feature Extraction

The BHSA Database

All Hebrew analysis uses the Biblia Hebraica Stuttgartensia Amstelodamensis (BHSA) database [16], accessed through Text-Fabric [17]. Extraction is restricted to words that BHSA marks as Hebrew, which matters more than it sounds: Ezra 4:8–6:18 and 7:12–26 are Aramaic and account for 28.9% of that book, and Jer 10:11 adds a further 19 words. Daniel's Aramaic section, 2:4b–7:28, is excluded by the chapter range assigned to that unit. BHSA supplies a complete morphological and syntactic annotation of the Masoretic text: for every word, a lexeme identifier, part of speech, verbal tense/aspect and stem, person, number, gender, pronominal suffix and state; and for every phrase and clause, a type, a syntactic function, and a relation to its governing clause. These annotations are the product of decades of work by the Eep Talstra Centre for Bible and Computer and are independent of any dating hypothesis.

The Anchored Corpus

The training corpus comprises 25 units spanning 760–167 BCE, listed in Table 1. Books with disputed internal composition histories are split into the conventional scholarly divisions (Isaiah into three, Zechariah into two), and the oracular core of Jeremiah is separated from its Deuteronomistic prose frame, which is treated as an undated target rather than an anchor.

The units differ in the kind of evidence that fixes their dates, and the difference is marked in the table along the lines set out above. Twelve carry an explicit synchronism and twelve a datable historical referent, neither of which depends on the feature cline under analysis. Jonah alone is dated principally by typology, following the critical commentary [18], with some recent work pressing it later still [19]; even there a terminus post quem is available, since Nineveh is described as a city that already lay in ruins (Jonah 3:3).

Table 2 reports how much any of this matters. A leave-one-unit-out jackknife, re-running the entire pipeline 25 times, gives a rank correlation between +0.48 and +0.76 with a median of +0.65, and pairwise accuracy between 66.4% and 78.6%. That spread is the leverage a single unit carries at this corpus size, and it falls almost entirely on the late anchors that fix the recent end of the scale: removing Nehemiah, Esther, Daniel, Ezra, or Jonah each costs between 0.14 and 0.17 of correlation, while removing any pre-exilic unit costs at most 0.05. The full-corpus figure of +0.66 should accordingly be read against that range rather than as a point estimate.

Removing the less securely anchored units changes little. Dropping all six that lack an explicit synchronism leaves ρ at +0.69 with a mean absolute error of 95 yr against a 127 yr baseline, better in both respects than the full corpus. Dropping Jonah and Ecclesiastes together costs 0.19 of correlation, which is what dropping any two late anchors costs and is not peculiar to them.

Why Passages Rather Than Books

The most consequential design choice in this article is that features are extracted from passages of about 500 words rather than from whole books.

Table 1. The training corpus. Dates are BCE; "Pass." is the number of ∼500-word passages. Type S dates rest on an explicit synchronism, a named ruler or regnal year checkable against Assyrian, Babylonian or Persian records. Type H dates rest on a datable event, institution, or language contact documented outside the biblical tradition. Neither depends on the feature cline this model estimates. Type T marks the one unit dated principally by resemblance to Hebrew of supposed date; Table 2 reports the effect of removing it.

Unit	Date	Words	Pass.	Type	Basis of the date
Amos	760	2,780	5	S	Superscription: Uzziah of Judah / Jeroboam II of Israel
Hosea	745	3,146	6	S	Superscription: Uzziah to Hezekiah / Jeroboam II
Micah	725	1,895	4	S	Superscription: Jotham, Ahaz, Hezekiah
Isaiah_1	720	13,504	26	S	Superscription: Uzziah to Hezekiah; Assyrian campaigns
Nahum	650	746	1	H	Fall of Nineveh, 612 BCE
Zephaniah	630	1,037	2	S	Superscription: Josiah
Habakkuk	605	897	2	H	Neo-Babylonian ascendancy
Jer_oracle	605	14,520	28	S	Superscription: Josiah to Zedekiah; 597 and 586 BCE
Obadiah	585	392	1	H	Edomite conduct at the fall of Jerusalem, 586 BCE
Ezekiel	580	26,182	51	S	Thirteen regnal dates from Jehoiachin's deportation, 597 BCE
Lamentations	570	1,945	4	H	Destruction of Jerusalem, 586 BCE
Isaiah_2	545	5,738	11	S	Cyrus named; fall of Babylon, 539 BCE
Haggai	520	877	2	S	Regnal dates, second year of Darius I, 520 BCE
Zechariah_1	518	2,484	5	S	Regnal dates, second to fourth year of Darius I
Malachi	460	1,187	2	H	Persian governor (*peha*); second-temple cult
Joel	400	1,318	3	H	Functioning second temple; no monarchy
Jonah	400	985	2	T	Nineveh already fallen (3:3); post-exilic novella
Isaiah_3	400	3,689	7	H	Restored temple cult; Persian-period institutions
Ezra	380	3,745	7	S	Persian regnal dates, Cyrus to Artaxerxes
Nehemiah	380	7,842	15	S	Twentieth year of Artaxerxes I, 445 BCE
Zechariah_2	350	1,987	4	H	Greek (*Yawan*) as a military power
Chronicles	330	35,330	69	H	Genealogies to the Persian period; Cyrus decree
Esther	300	4,621	9	H	Achaemenid court and administration; Xerxes
Ecclesiastes	250	4,233	8	H	Persian loanwords (*pardes*, *pitgam*)
Daniel	167	3,437	7	H	Antiochene persecution, 167 BCE

At the level of whole books the corpus offers only 25 observations. Any feature set large enough to describe Hebrew morphosyntax then has more dimensions than there are data points. The standard responses to this — screening features by their correlation with date, or reducing dimension using the date labels — both consult the labels before cross-validation and thereby inflate apparent performance. Worse, a whole-book rate is a single number summarizing tens of thousands of words, so a book whose passages are internally consistent cannot be told apart from one that is a composite.

Partitioning at ∼500 words yields 281 passages from the same 144,517 words. This is not a free multiplication of the sample size. Passages from one book are correlated, so the appropriate measure is the effective sample size $n_{\text{eff}} = n/(1 + (\bar{m} - 1)\rho_I)$, where ρ_I is the intraclass correlation and $\bar{m}$ the mean number of passages per book. Across passage sizes from 200 to 1000 words the median intraclass correlation runs

Table 2. Robustness of the headline statistics. MAE and baseline are in years, the baseline being the constant predictor on the same subset. The jackknife drops each anchored unit in turn and re-runs the entire pipeline on the remaining 24; its spread is the leverage any single unit carries at this corpus size, and the full-corpus ρ should be read against it rather than as a point estimate. The lower block removes the units whose dates are least independent of the language: Jonah, the one unit dated principally on typological grounds, together with Ecclesiastes, and then all six units lacking an explicit synchronism.

Fit	**Units**	**MAE**	**Baseline**	ρ	**Pairwise**
Full corpus	25	118	137	+0.66	72.9%
Leave-one-unit-out jackknife, all 25 fits					
median	24	—	—	+0.65	72.3%
range	24	—	—	+0.48 to +0.76	66.4–78.6%
Removing the less securely anchored units					
without Jonah and Ecclesiastes	23	120	130	+0.51	66.8%
without all 6 non-synchronism units	19	95	127	+0.69	74.6%

0.20–0.32, and the effective sample size is three to seven times the book-level figure (S2 Table). That is the real gain, and it is what makes a feature set of this width estimable at all.

Passage size trades two effects against each other. Smaller passages give more observations but sparser features, since a lexical item occurring once per thousand words will be absent from most 200-word passages, and its rate is then not a measurement but a coin flip. Larger passages give denser features but fewer observations. I therefore crossed passage sizes of 200, 300, 400, 500, 700, and 1000 words with three feature sets and with calibration on and off, 36 configurations in all, selecting on out-of-sample rank correlation under leave-one-book-out (S3 Table). The selected configuration uses ~500-word passages, though performance is flat between 400 and 1000, so the choice is not a delicate one.

Passages respect verse boundaries: words are accumulated verse by verse until the target is reached. A trailing remainder shorter than half the target is merged into the preceding passage rather than kept as a short, high-variance observation.

Feature Extraction

The feature set is fixed by frequency rank computed over the whole BHSA corpus, without reference to date. 634 features are extracted, of which 56 are structurally constant on this corpus and are dropped inside each fold, leaving 578 in six families:

1. **Lexical rates** (250 features): occurrences per 1000 words of each of the 250 most frequent lexemes in the corpus. Since selection goes by corpus frequency, these are dominated by function words — articles, prepositions, conjunctions, pronouns — rather than by the rare diagnostic vocabulary of the CBH/LBH literature. This is deliberate, as the sparsity constraint above rules out the rare items, and the consequences of that are taken up below.

2. **Part-of-speech transition rates** (207 features): rates of each ordered pair of parts of speech across adjacent words, capturing local syntactic profile without reference to any particular construction.

3. **Verb morphology** (57 features): distributions over tense/aspect (qatal, yiqtol, wayyiqtol, participle, infinitive, imperative), stem (qal, nifal, piel, hifil, hitpael, pual, hofal), and their interaction.

4. **Phrase and clause structure** (40 features): phrase type and syntactic function rates, and clause-relation rates from the BHSA clause hierarchy.

5. **Agreement and suffix morphology** (19 features): person, number, gender and pronominal-suffix rates.

6. **Structural** (5 features): mean phrase and clause length, clauses per passage, and type–token ratio.

Two extensions were built and evaluated and neither is used. A character and part-of-speech n-gram extension adding 689 further features does not improve out-of-sample performance at the selected passage size (S3 Table). A clause-type family — rates and transition rates over the BHSA clause types that encode the Hebrew verbal system, including the wayyiqtol narrative clause (Way0) and the weqatal clause (WQt0) — was added after an audit found that an earlier implementation had counted clause types while indexing them by clause relation, whose value sets are disjoint, so that the family had been silently absent. Implemented correctly, its members are among the most strongly date-correlated single features in the corpus, reaching $|\rho| = 0.54$ for fronted yiqtol and 0.50 for weqatal clauses, and yet adding all 50 of them leaves out-of-sample performance unchanged (ρ +0.665 to +0.660, mean absolute error 118 to 120 yr). Their information is already carried by the verb-morphology and part-of-speech transition features. The reported model is therefore the one without them, and the exercise is reported because it indicates the feature set is saturated rather than arbitrary.

Features whose standard deviation is zero on the training fold, or which are missing in more than 20% of passages, are dropped inside the fold. Remaining missing values are imputed at the training-fold median.

Model

Weighted Ridge Regression

Let $\mathbf{X}$ be the $n \times p$ matrix of passage features and $\mathbf{y}$ the vector of book dates, each passage inheriting its book's date. Features are centered and scaled on the training fold. The estimator is ridge regression,

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \sum_{i=1}^{n} w_i \left(y_i - \mathbf{x}_i^\top \boldsymbol{\beta}\right)^2 + \lambda \|\boldsymbol{\beta}\|^2, \tag{1}$$

solved in dual form, $\hat{\boldsymbol{\beta}} = \mathbf{A}^\top (\mathbf{A}\mathbf{A}^\top + \lambda \mathbf{I})^{-1} \tilde{\mathbf{y}}$, where $\mathbf{A}$ is the weighted standardized design. Because $p > n$ on the Hebrew side, the dual form is the cheaper route, and one eigendecomposition of $\mathbf{A}\mathbf{A}^\top$ serves the entire λ sweep. λ is selected by an inner leave-one-book-out loop nested inside each outer fold, never using the held-out book.

Inverse-Density Weighting

The anchored corpus is not uniform in time. Post-exilic material outnumbers pre-exilic material, and an unweighted fit is pulled toward the dense region: in the unweighted model no book receives a confidently pre-exilic estimate, which is a property of the sampling rather than of Hebrew. Weights are therefore taken inversely proportional to a kernel estimate of the date density,

$$w_i \propto \left[\sum_j \exp\left(-\tfrac{1}{2} \left(\frac{d_j - d_i}{h} \right)^2 \right) \right]^{-\alpha}, \tag{2}$$

with bandwidth $h = 90$ yr and exponent $\alpha = 1$, normalized to mean one. Weighted ridge is implemented as ordinary ridge on rows scaled by $\sqrt{w_i}$.

Variance Matching and Its Cost

A regularized regression is a shrinkage estimator, so its predictions have systematically smaller variance than the quantity being predicted. Under plain ridge, the predicted dates for the anchored corpus span roughly a third of the true range, and no text can then be assigned a confidently early or confidently late date. This is an artifact of shrinkage rather than a finding about Hebrew. The correction is to match the first two moments of the out-of-sample predictions to the true dates,

$$\hat{d} \;=\; \bar{d} \;+\; S\,(\hat{d}_{\mathrm{raw}} - \bar{d}_{\mathrm{raw}}), \qquad S = \frac{\mathrm{sd}(d)}{\mathrm{sd}(\hat{d}_{\mathrm{raw}})}, \tag{3}$$

with $\bar{d}$, $\mathrm{sd}(d)$, $\bar{d}_{\mathrm{raw}}$ and S all computed from leave-one-book-out predictions on the training corpus only. Here $S = 1.79$. The step is the inverse-calibration correction familiar from the classical-versus-inverse regression literature [20]: when the object is to recover the predictor rather than to minimise squared error in the response, matching the spread of the estimates to the spread of the quantity being estimated is the appropriate adjustment.

This is a deliberate trade, and both sides of it ought to be stated. Rescaling *increases* the mean absolute error, since inflating the spread of the predictions amplifies their errors along with their signal: the uncalibrated model gives 94 yr against 118 yr for the rescaled one. What the trade buys is a model that is two-sided. Uncalibrated predictions span only 294 yr against 593 yr of real dates and place just 4 of the 8 genuinely pre-exilic books before the exile. After rescaling, the predictions span 526 yr and 6 of 8 are placed correctly.

Because the substantive question in the Pentateuchal debate is which side of the exile a text falls on, a model that can hardly ever say "early" is useless for that purpose whatever its mean absolute error. Its inability to say "early" is a property of the estimator, not evidence that nothing is early. Readers whose interest is point accuracy rather than calibration should use the uncalibrated variant, whose figures are in S3 Table.

Conformal Prediction Intervals

Intervals are conformal, calibrated on the absolute leave-one-book-out residuals of the rescaled predictions. For coverage level $1-\alpha$ the half-width is the $\lceil (n+1)(1-\alpha) \rceil$-th smallest absolute residual. This is distribution-free and has finite-sample marginal coverage [21, 22]: it assumes only that the target text is exchangeable with the anchored corpus, not that residuals are Gaussian or homoscedastic. On this corpus the half-widths are ± 137 yr at 68% and ± 281 yr at 90%.

For a target unit, $P(\text{post-exilic})$ is computed by adding the empirical residual distribution to the point estimate and taking the fraction of the resulting draws that fall after 586 BCE. This inherits the conformal guarantee and makes no distributional assumption.

Validation

A Failure Mode That Inflates Reported Accuracy

Reported accuracies for linguistic dating models of this class vary by two orders of magnitude. Much of that variation is not a difference in the underlying signal but an artifact of how the models are validated, and the artifact is worth naming because it is easy to commit and hard to see.

In a Bayesian dating model, the posterior over a held-out text's date combines a likelihood from the linguistic features with a prior. Should the prior for the held-out text be centered on that text's own scholarly date, which is the natural thing to do when the prior is meant to encode what scholars think, then the model is being asked to recover a number it has already been handed. The resulting holdout error measures the width of the prior rather than the information in the language.

The effect is large. Under a prior-centered design on this corpus, the median share of posterior precision contributed by the linguistic evidence is 11.7%, and mean holdout error is 44 yr. Under an agnostic prior with an otherwise identical pipeline, the same model achieves 156 yr. For the 8 units whose scholarly dates carry the tightest uncertainty ($\sigma \leq 20$ yr), the linguistic evidence supplies 4.5% of posterior precision and the leaky design reports 6.6 yr against an honest 189 yr (Table 3). For a model of this kind, a reported accuracy of a decade or two is a signature of the design rather than an achievement.

All results below use an agnostic prior; the reported model is not Bayesian and has no prior to leak.

The Leave-One-Book-Out Protocol

Cross-validation folds are whole books, never passages. Passages from one book are strongly correlated, so splitting them across a fold boundary would let the model see the held-out book's own material. Within each fold, and using only the training books, features with zero variance or excessive missingness are dropped,

Table 3. Prior leakage inflates reported holdout accuracy. "Prior σ" is the standard deviation of the scholarly prior in years; "data share" is the fraction of posterior precision contributed by the linguistic features; the two error columns are mean absolute errors in years under a prior centered on the unit's own scholarly date and under an agnostic prior, with the pipeline otherwise identical.

Unit	Prior σ	Data share	Leaky err.	Honest err.
Haggai	10	2.3%	6	334
Daniel	10	1.7%	3	233
Zechariah_1	10	1.5%	4	236
Amos	20	5.8%	8	144
Habakkuk	20	5.4%	3	51
Isaiah_2	20	8.0%	15	267
Ezekiel	20	5.9%	10	158
Zephaniah	20	5.6%	4	90
Malachi	25	8.9%	18	192
Micah	25	9.7%	3	39
Nahum	25	8.6%	4	48
Hosea	25	9.2%	1	5
Isaiah_1	30	12.0%	16	140
Jer_oracle	30	11.7%	11	105
Chronicles	40	21.8%	20	116
Lamentations	50	35.8%	64	238
Jonah	60	34.1%	18	50
Ezra	60	33.8%	26	80
Esther	60	64.0%	80	200
Nehemiah	60	34.5%	36	116
Obadiah	80	49.2%	29	57
Isaiah_3	100	63.3%	186	296
Ecclesiastes	120	70.7%	150	212
Joel	150	78.3%	200	256
Zechariah_2	150	78.5%	190	242
Median, all	30	11.7%	16	144
Mean, $\sigma \leq 20$ (8 units)	16	4.5%	6.6	189.1

medians for imputation are computed, standardization constants are computed, inverse-density weights are computed, λ is selected by an inner leave-one-book-out loop, and the variance-matching constants and conformal residuals are derived. The held-out book's date is used only to score the result.

A book's estimate is the median of its passage estimates. The dispersion of those passage estimates is reported separately (Table 4) and is informative in its own right, being small for compositionally uniform books and large for composites.

What the Signal Supports

Under this protocol the model achieves a mean absolute error of 118 yr, against 137 yr for the constant predictor that assigns every book the corpus mean date. Rank correlation between true and estimated date is $+0.66$ ($p = 0.0003$), and 72.9% of the 295 book pairs are placed in the correct chronological order.

However, the same corpus was used to choose the passage size and feature set, so parametric p-values would be optimistic. Significance is therefore established by a permutation null in which book dates are shuffled and the *entire* pipeline is re-run — weighting, λ selection, variance matching and all — 300 times. Permutation is at book level, not passage level, so the clustering structure is preserved under the null; a passage-level shuffle would break the correlation between passages of the same book and produce a null that is far too easy to beat. Against this null the observed rank correlation of $+0.66$ has $p = 0.003$ (null median -0.03), the pairwise ordering accuracy of 72.9% has $p = 0.013$ (null median 50.0%), and the mean absolute error of 118 yr has $p = 0.003$ (null median 179 yr).

One arithmetic point deserves emphasis, since pairwise ordering accuracy is commonly reported in this literature without it. The 295 pairs formed from 25 books are not independent, as each book appears in $25-1$ of them. Treating the accuracy as a binomial proportion over 295 independent trials gives p-values wrong by ten orders of magnitude. The permutation figures above are the correct ones, and any comparison

Table 4. Leave-one-book-out performance. Dates BCE; error is true minus estimated, so a positive error means the model placed the book too late. "Pass. SD" is the standard deviation of the book's passage-level estimates, a measure of internal homogeneity. Every quantity is out-of-sample: standardization, weighting, λ selection and calibration were performed without the held-out book.

Unit	True	Est.	Err.	Pass. SD	P(post-ex.)
Amos	760	623	+137	40	0.44
Hosea	745	563	+182	85	0.60
Micah	725	626	+99	67	0.44
Isaiah_1	720	439	+281	95	0.84
Nahum	650	684	-34	0	0.28
Zephaniah	630	661	-31	118	0.32
Habakkuk	605	685	-80	43	0.28
Jer_oracle	605	670	-65	80	0.28
Obadiah	585	625	-40	0	0.44
Ezekiel	580	469	+111	154	0.76
Lamentations	570	696	-126	68	0.28
Isaiah_2	545	667	-122	71	0.28
Haggai	520	551	-31	111	0.60
Zechariah_1	518	431	+87	52	0.84
Malachi	460	515	-55	86	0.64
Joel	400	566	-166	109	0.60
Jonah	400	349	+51	10	0.96
Isaiah_3	400	515	-115	72	0.64
Ezra	380	184	+196	55	1.00
Nehemiah	380	188	+192	83	1.00
Zechariah_2	350	544	-194	49	0.60
Chronicles	330	324	+6	110	0.96
Esther	300	171	+129	54	1.00
Ecclesiastes	250	379	-129	120	0.96
Daniel	167	449	-282	104	0.80

with pairwise accuracies reported elsewhere should take this into account.

Correcting for the Configuration Search

The figures above come from a configuration — passage size, weighting exponent, ridge penalty — that was itself chosen by comparing out-of-sample performance across a search on this same corpus. A permutation null that holds the winning configuration fixed does not account for that choice, so the reported correlation is the maximum over a search and is optimistic by an unknown amount.

The amount can be measured by nesting the search. For each held-out book, the entire configuration was selected by an inner leave-one-book-out loop over the remaining 24 units, drawn from a grid of 36 combinations, and only then was the held-out book predicted. Nothing about the configuration is informed by the book being scored.

Under that protocol the mean absolute error is 124 yr against 137 yr for the constant predictor, the rank correlation is +0.62 ($p = 0.0009$), pairwise ordering is 71.5%, and 18 of 25 books fall on the correct side of the exile. Selection therefore inflates the rank correlation by about 0.06 and the pairwise accuracy by roughly two points. The attenuated figures are the ones a fresh corpus should be expected to reproduce, and the substantive conclusions do not depend on the difference. The search is also stable: the same weighting exponent is chosen in 24 of the 25 folds, and the passage size alternates between the two adjacent settings whose performance the sweep found indistinguishable.

Variance matching is excluded from this grid deliberately. It is a positive linear map of the predictions, so it leaves both the rank correlation and the pairwise ordering exactly unchanged and cannot inflate either. It is applied always, as an a priori correction for shrinkage of the kind long used in inverse calibration [20], rather than tuned for performance.

Two-Sidedness and Calibration

What matters for the Pentateuchal application is whether the model can say “early” as well as “late”. After inverse-density weighting and variance matching, 6 of the 8 genuinely pre-exilic books are placed before 586 BCE, and 14 of the 17 post-exilic ones after it, giving 20 of 25 overall (Table 4). Without those two steps the model places almost nothing before the exile, and its apparent confidence that everything is late is an artifact of how the corpus is distributed in time rather than evidence about any text.

Fig 1 shows estimated against true date with the conformal bands. Empirical coverage of the nominal 68% band is slightly conservative, which is the expected behavior of conformal intervals at this sample size.

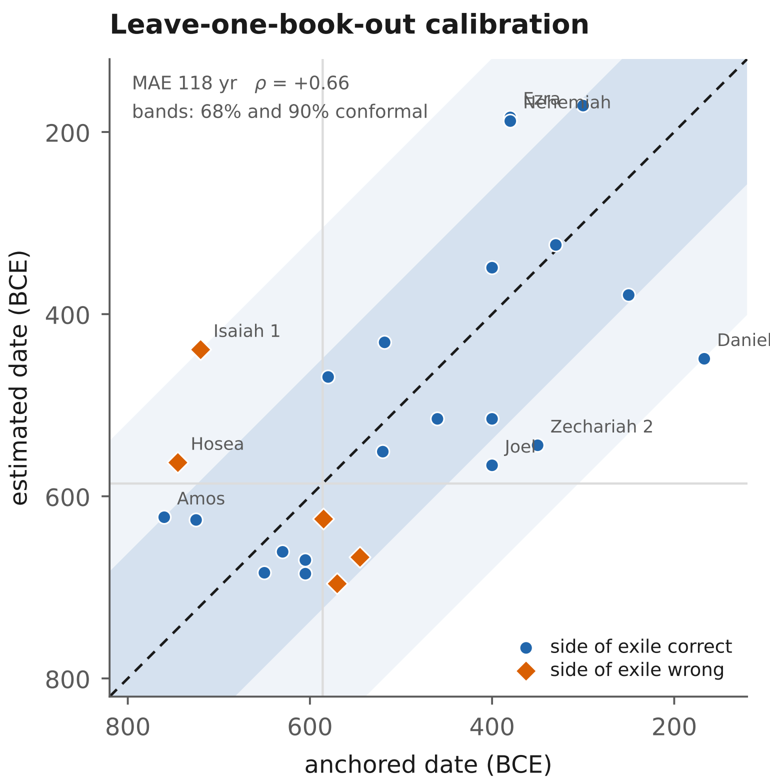

Fig 1. Leave-one-book-out calibration. Each point is one anchored book, dated by a model fitted without it. The dashed line is exact recovery; shaded bands are the 68% and 90% conformal intervals. Books whose estimate falls on the wrong side of the exile are marked with diamonds. Standardization, weighting, regularization and interval calibration were all performed inside the fold.

What the Signal Does Not Support

Three limits ought to be stated in the same breath as the results.

First, 118 yr against a 137 yr baseline is a real improvement but a modest one. What the model provides is a chronological *ordering* that has been calibrated onto an absolute scale, not a precise clock. The reported intervals of ±137 yr at 68% express as much.

Second, the largest single error is Daniel at 282 yr. Genre drives most of this, as narrative books are placed late and poetic books early relative to their true dates (see below). Any single estimate for a text of unusual genre should be read accordingly.

Third, nothing here resolves dates *within* the post-exilic period. A text estimated at 380 BCE with a ±137 yr interval cannot be distinguished from one estimated at 300 BCE. The claims that follow are therefore confined to the pre-exilic/post-exilic distinction and to relative ordering, both of which the validation supports.

Results: Dating the Undated Units

The model, fitted on all 25 anchored books, was applied to 19 undated units: the three classical documentary sources, their conventional sub-strata, the five Pentateuchal books as received, the Deuteronomistic prose of Jeremiah, and the three archaic poems. Source partitions follow the standard critical division at chapter granularity, specified in S1 Appendix. Chapter granularity is exact for the sub-strata, whose ranges partition their parent sources without remainder, and nearly so for the three sources: Genesis 6 and 25, where the Priestly and JE material is interwoven verse by verse, are the only chapters assigned to two sources. No target contributed to training, feature selection, weighting, regularization or calibration.

Table 5. Undated units. Dates BCE; intervals are 68% conformal, calibrated on leave-one-book-out residuals. No unit here contributed to training, feature selection, weighting, regularization or calibration. The three poems are extracted at verse precision (see text).

Unit	Words	Pass.	Est.	68% interval	P(post-ex.)
Archaic poems					
Song of Deborah	462	1	811	948–673	0.04
Song of the Sea	433	1	638	776–501	0.40
Song of Moses	782	2	524	661–386	0.64
Documentary sources					
JE composite	37,750	74	418	556–281	0.84
Sub-strata					
Genesis JE	22,407	44	432	569–294	0.84
Exodus JE	11,780	23	361	498–224	0.96
Numbers JE	3,563	7	443	580–305	0.84
D source	20,128	39	372	509–234	0.96
D law code	7,353	14	422	559–284	0.84
D frame	11,993	23	324	462–187	0.96
P source	54,435	106	340	477–202	0.96
Leviticus P	10,496	21	311	448–174	0.96
Holiness Code	5,958	12	390	528–253	0.96
Jeremiah Dtr prose	15,197	30	446	584–309	0.84
Pentateuch as received					
Genesis	28,762	56	419	556–281	0.84
Exodus	23,748	47	355	492–217	0.96
Leviticus	17,099	33	334	471–197	0.96
Numbers	23,188	45	353	490–215	0.96
Deuteronomy	20,128	39	372	509–234	0.96

The Documentary Sources Are Post-Exilic

The Priestly source is placed at 340 BCE (68% conformal interval 477–202 BCE) with $P(\text{post-exilic}) = 0.96$; the Deuteronomic source at 372 BCE (509–234) with 0.96; and the JE composite at 418 BCE (556–281) with 0.84 (Fig 2). All three fall after the exile with probability at least 0.84.

The relative ordering is JE $\rightarrow$ D $\rightarrow$ P, the classical sequence of the documentary hypothesis, recovered from linguistic evidence alone by a model trained only on independently anchored prophetic and post-exilic books. The *ordering* is the better-supported half of this result. With intervals of 137 yr, the ranges of the three sources overlap substantially, and the ordering claim rests on the model's demonstrated ability to order rather than on any separation between the point estimates.

Sub-Strata

Splitting Deuteronomy into its law code (chapters 12–26) and its narrative frame places the code at 422 BCE and the frame at 324 BCE — the code earlier than the frame, in the direction the critical consensus expects if the code is the older core around which the frame was later built. Within Leviticus, the Holiness Code (17–26) is placed at 390 BCE against 311 BCE for the Priestly material of 1–16. The JE material of Genesis,

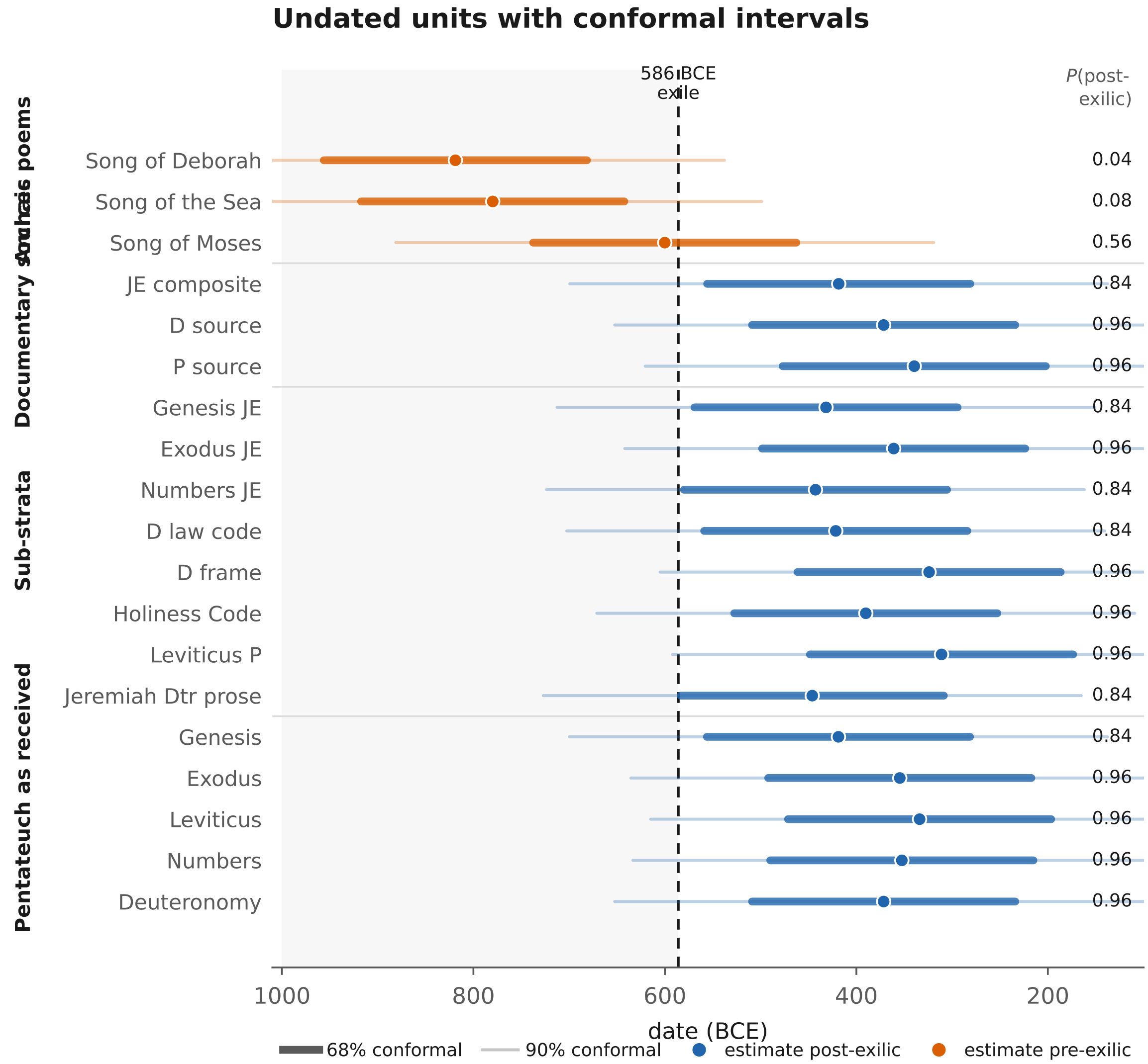

Fig 2. Undated units with conformal intervals. Thick bars are 68% intervals, thin bars 90%; the dashed line is the exile of 586 BCE. Colour marks which side of the exile the point estimate falls on. The three poems are extracted at verse precision. No unit shown contributed to training, feature selection, weighting, regularization or calibration.

Exodus and Numbers is placed at 432, 361 and 443 BCE respectively. The Deuteronomistic prose of Jeremiah is placed at 446 BCE.

These differences among sub-strata are small relative to the interval width. They ought to be read as ordering statements consistent with existing literary-critical arguments, not as independent confirmation of them.

The Archaic Poems and an Extraction Artifact

The three poems traditionally regarded as the oldest Hebrew verse [10, 23, 24] needed a correction before they could be dated at all.

Poems in this corpus are conventionally identified by chapter, but the chapters containing them also contain prose. Exodus 15 is 47% prose by word count, and Deuteronomy 32 is 26%. Because narrative is systematically placed late by this model, including the prose frame drags the poem's estimate later. The effect is not small. The Song of the Sea moves from 638 BCE when extracted as a whole chapter to 780 BCE when extracted at verse precision (Exod 15:1–18). The verse-precise figures are the ones reported here; the chapter-level ones are an artifact, and the convention that produces them is widespread.

At verse precision the Song of the Sea is placed at 780 BCE (917–642, $P(\text{post-exilic}) = 0.08$), and the Song of Deborah at 819 BCE (956–681, 0.04). These are the two earliest of the 19 undated units, agreeing with a century of philological argument that they are the oldest surviving Hebrew, and arrived at without any training on them.

The prose continuation of the Song of the Sea (Exod 15:19–27) is placed at 469 BCE, 311 yr later than the poem it follows. A gap of that size between adjacent verses is itself evidence that the poem is not a composition of its narrative frame. Whether such a gap could have been manufactured is taken up below.

The Song of Moses (Deut 32:1–43) is placed at 600 BCE (738–463, 0.56), appreciably later than the other two and straddling the exile. A reading of its contents would not predict this. Its divine-council theology, in which Elyon apportions the nations among the gods and YHWH receives Israel as his share (32:8–9), is usually taken as among the most archaic material in the Hebrew Bible. But the model sees the language rather than the theology, and the language is late. Whether the archaic theology of these verses was preserved in a later idiom, or whether the linguistic estimate is simply wrong for a text this short, cannot be settled from the present evidence.

Genre as the Dominant Confound

Residuals on the anchored corpus are structured by genre, with narrative books placed *late* and poetic books *early* relative to their true dates. The clearest case is Lamentations, whose date is fixed by its subject matter, since it laments the destruction of 586 BCE, making it as securely anchored as any text in the corpus. The model places it at 696 BCE against a true 570 BCE, an error of 126 yr toward greater antiquity. Lamentations is poetry.

The pattern across the corpus is consistent: narrative books carry a mean residual of +115 yr, poetry -126, and the 17 prophetic books, which dominate the corpus, only -2.1. Genre displaces an estimate by roughly the width of a conformal interval.

This bears directly on the poems above. If poetry as such is placed early by this model, then some part of the antiquity assigned to the Songs of the Sea and Deborah is genre rather than date. I decline to apply a correction, since with one securely dated poetic book any such correction rests on a single observation. What can be done instead is to ask whether the conclusion survives applying it anyway. Subtracting the full 126 yr poetic residual moves the Song of the Sea from 780 to 654 BCE and the Song of Deborah from 819 to 693 BCE. Both remain pre-exilic, both remain the earliest of the 19 undated units, and their probability of being post-exilic rises only to 0.32 and 0.28. For the Song of the Sea to cross the exile, the genre effect would have to be 1.5 times the value observed in Lamentations.

The early placement of the poems is therefore partly confounded with genre and survives the confound at its observed magnitude. Enlarging the anchored poetic corpus remains the way to separate the two properly.

The Price of Archaizing

Everything above depends on one premise: that the language of a text reflects when it was written rather than when its author wished to seem to be writing. The size of the gap between those two things is measured here, first by performing archaizing on Hebrew directly, and then by turning to Greek texts whose authors performed it themselves.

Synthetic Archaizing in Hebrew

Take a securely dated *late* book, replace its LBH forms with their CBH counterparts at a controlled rate r, re-extract all 578 features from the modified text, and re-date it with a model trained on the other 24 books. Since the substitution was performed by the investigator, the ground truth is known exactly.

The substitutions are the classical diagnostics of the CBH/LBH literature, applied in the LBH → CBH direction: the relative particle *še-* → *ašer*; the first-person pronoun *anī* → *ankī*; *malkūt* → *mamlākāh*; and the negator *'ēn* → *lō'*, which changes part of speech as well as lexeme. Substitution is applied at the token level before feature extraction, so every downstream feature — lexeme rates, part-of-speech transitions, phrase types, clause relations — sees the altered text exactly as it would if a scribe had written it that way. The rate r runs from 0 to 1, and at $r = 1$ not one LBH token of these types survives anywhere in the book.

Table 6. What total lexical archaizing buys. Each securely dated late book had its Late Biblical Hebrew forms replaced by Classical counterparts at rate r and was then re-dated by a model trained on the other 24 books. $r = 1$ replaces every eligible token in the book. Dates BCE; a positive shift means the book appears older. Ezra, Nehemiah and Chronicles offer too few eligible tokens for their null results to be informative.

Book	**True**	$r = 0$	$r = 1$	**Shift**	**Swaps**	**per 1k w**
Chronicles	330	298	296	-1	94	2.7
Esther	300	134	139	+5	42	9.1
Ecclesiastes	250	393	424	+30	142	33.5
Daniel	167	441	463	+22	47	13.7
Ezra	380	145	145	+0	13	3.5
Nehemiah	380	155	150	-5	28	3.6
Mean apparent shift				**+9**		

Total lexical archaizing moves the estimated date by a mean of +9 yr, with a maximum of +30 yr across the 6 books (Table 6). The response is gradual rather than thresholded, as there is no substitution rate at which the estimate suddenly jumps.

However, three of the six books had almost nothing to substitute, with Ezra offering 13 eligible tokens in the entire book, so their null results carry little information. One cannot archaize a text that already lacks the forms. Among the 3 books with substantial substitution density, the exchange rate is about 1.0 yr of apparent antiquity per substitution per 1000 words. Ecclesiastes at $r = 1$ has 33.5 substitutions per 1000 words, roughly one word in thirty replaced, which is a density no scribe sustains across a book, and it moves +30 yr.

Where the Model's Leverage Lies

The reason for so small a number is visible in the model's coefficients.

Fig 4 gives the change in estimated date produced by a one-standard-deviation shift in each feature. Lexical features carry 47% of the total leverage, but that share is spread across 250 features, none of which is worth more than 12.1 yr per standard deviation. No single word is a lever. The rest of the leverage sits in part-of-speech transition rates, verbal aspect and stem distributions, and clause-relation profiles, which are high-frequency grammatical habits rather than choices of vocabulary.

Moreover, these are the features that the passage-level design forces the model to rely on. Requiring a feature to be measurable within 500 words demotes the rare diagnostic vocabulary, which is the manipulable part of the language, and promotes high-frequency morphology, which is not. The resistance to archaizing documented below is therefore a structural consequence of the passage-level design rather than an additional assumption: any model constrained to features estimable in a few hundred words will rest on the same high-frequency morphology.

This can be put quantitatively. Because the model is linear in standardized features, the smallest coordinated feature shift that displaces an estimate by D years has length $D/\|\boldsymbol{\beta}\|$ in standard deviations. For the 311 yr gap between the Song of the Sea and its prose frame, that comes to 5.0 standard deviations across 578 dimensions. What is required is not a word swap but a systematic re-profiling of a text's morphology and syntax, sustained across the whole passage. As the Greek results below show, this is achievable by authors trained to achieve it.

Greek Archaizing That Really Happened

The Hebrew experiment simulates archaizing, and it simulates only the lexical component of it. Ancient Greek supplies the real thing, already labeled. During the Second Sophistic, prose authors of the first three centuries CE deliberately wrote in imitation of Classical Attic, four to seven centuries after the fact. The

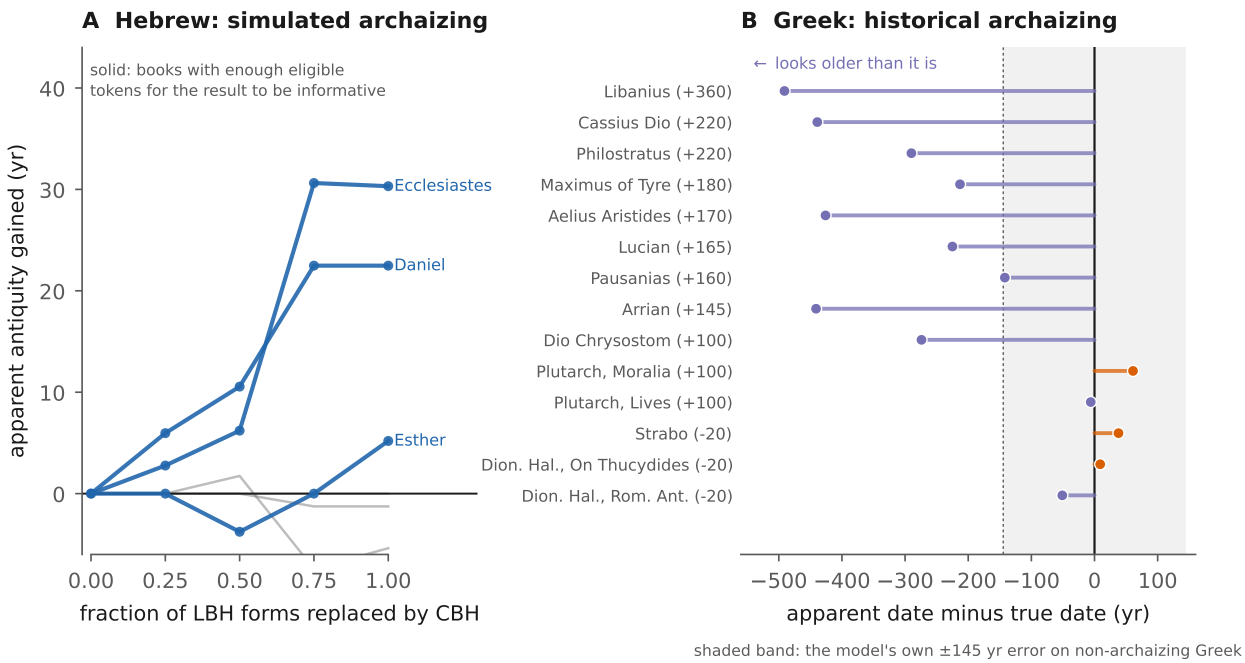

Fig 3. The price of archaizing, by mechanism. A, Hebrew: apparent antiquity gained as Late Biblical Hebrew forms are replaced by Classical counterparts at increasing rates in six securely dated late books. Solid lines are the books with enough eligible tokens for the result to be informative. **B**, Greek: displacement of the Second Sophistic Atticizers under a model trained only on non-archaizing Greek, with the model's own error on non-archaizing texts shaded for comparison. Lexical substitution moves an estimate by tens of years; reconstructed grammar moves it by hundreds.

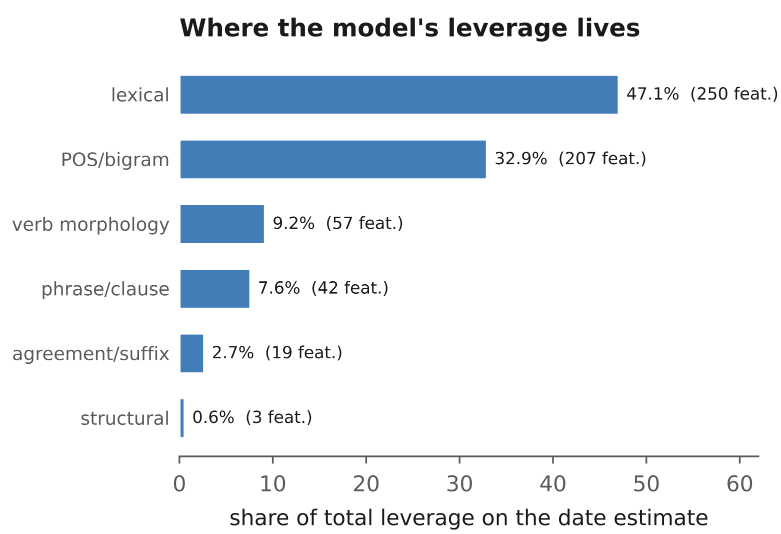

Fig 4. Where the model's leverage lives. Share of total leverage on the date estimate by feature family, where leverage is the absolute change in estimated date produced by a one-standard-deviation shift in a feature, summed within family. Lexical features carry the largest share but spread it across 250 weak features; the majority of leverage sits in morphology and syntax.

program was explicit and institutional. Attic prose composition was taught in the rhetorical schools, dictionaries of permitted Attic usage were compiled, and the imitation succeeded well enough that its practitioners were read as models of style for a millennium. Their dates of composition are securely known from external evidence. This is the labeled set of archaizing texts that Hebrew cannot provide.

The same pipeline was built on a 42-text corpus of non-archaizing Greek prose spanning the Classical and Koine periods, partitioned into 5116 passages, with 747 features. The texts were collected from the Perseus and First1KGreek repositories, where an author's corpus is stored one work per file, so that the orators, the Lives of Plutarch and the dialogues of Lucian each occupy dozens of separate files. Collecting only the complete corpus of each author matters: an earlier version of this corpus resolved a single work per author and thereby represented Lysias by one speech and Plutarch by one Life, which left the Classical end of the scale, the part the model most needs, severely under-sampled. Greek lacks a morphological annotation comparable to BHSA, so hand-built morphosyntactic rates (optative, infinitive and participle rates, particle and negation rates, the $\tau\tau$ versus $\sigma\sigma$ contrast, sentence length) are supplemented by word-form, character n-gram, and function-word bigram rates, all of them high-frequency by construction as on the Hebrew side. Under leave-one-text-out, the Greek model gives 145 yr against 238 yr for the constant predictor, a rank correlation of +0.69, and 74.7% pairwise ordering accuracy. The replication is therefore a real one, with the same design recovering a diachronic signal in an unrelated language.

The 14 Atticizing texts were excluded from training entirely and then dated.

Table 7. Second Sophistic Atticizers under a model trained only on non-archaizing Greek. Dates CE (negative = BCE). A negative shift means the text is dated earlier than it was written, i.e. the archaizing worked. These 14 texts were excluded from training entirely.

Author	**Work**	**True**	**Est.**	**Shift**	**Pass.**
Dionysius of Halicarnassus	Roman Antiquities	-20	-71	-51	500
Dionysius of Halicarnassus	Critical essays (15 works)	-20	-11	+9	108
Strabo	Geography	-20	+18	+38	268
Plutarch	Parallel Lives (87 works)	+100	+94	-6	419
Plutarch	Moralia (58 works)	+100	+161	+61	225
Dio Chrysostom	Orations (selected)	+100	-174	-274	308
Arrian	Anabasis of Alexander	+145	-296	-441	140
Pausanias	Description of Greece	+160	+18	-142	223
Lucian	Complete works (71 works)	+165	-60	-225	446
Aelius Aristides	Orations and Sacred Tales (56 works)	+170	-256	-426	540
Maximus of Tyre	Orations (Dialexeis, 5 works)	+180	-33	-213	19
Philostratus	Lives of the Sophists	+220	-70	-290	42
Cassius Dio	Roman History (epitome books)	+220	-219	-439	262
Libanius	Orations (selected)	+360	-131	-491	216
Mean displacement				**-206**	

The mean displacement is -206 yr, with 11 of the 14 texts placed earlier than they were written (Fig 3B). Libanius is displaced -491 yr. Set against the model's own 145 yr error on non-archaizing texts, this is a large effect. Deliberate and professionally trained archaizing displaces a text by more than a full error width, and in the worst cases by half a millennium.

The displacement also tracks how programmatic an author's Atticism was, which is the pattern one would expect if it measures imitation rather than noise. The strict Atticists are displaced furthest: Libanius by -491 yr, Arrian by 441, Cassius Dio by 439, Aelius Aristides by 426, Philostratus by 290, Dio Chrysostom by 274, and Lucian by 225. The authors who wrote a literary Koine rather than a reconstructed Attic are barely moved at all: Plutarch's *Lives* by 6 yr, Strabo by +38, and Dionysius' critical essays by +9. A dose-response relation of this kind across authors whose Atticizing intensity is independently described in the literary histories is difficult to obtain from a model that is merely noisy.

Two features of the Greek corpus bear on how this number should be read.

First, the effect is if anything *understated*, because the training set is contaminated. The corpus manifest classifies Aelian, Appian, Diogenes Laertius, Plotinus, Porphyry, Iamblichus, and the sophistic novelists as Koine, so all of them were used as non-archaizing training material. Yet every one is described in the literary-historical record as classicizing, and the model placed them among the largest early displacements in the training set even after being told they were not archaizing. A training set containing archaizers teaches the model that late texts look early, which shrinks the measured displacement of the declared Atticizers. Since the classification is a scholarly judgment rather than a datum, I do not adopt one version of it and defend it. Instead the measurement is reported under three definitions of the training set: the manifest's own labels, a stricter classification that reclassifies the authors named above, and a date-blind variant that discards register labels entirely and trains only on texts composed before 50 CE, prior to the Second Sophistic (S6 Table).

The result holds in sign and grows under the stricter definitions. The mean displacement is -206 yr under the manifest's labels, -297 yr under the stricter classification, and -397 yr under the date-blind variant, with 44 of 47 individual texts across the three placed earlier than they were written. The stricter variant is worth a further remark. It removes eight texts from training, and yet the model's own leave-one-text-out error *falls* from 145 yr to 101 yr while its rank correlation rises from +0.69 to +0.80. Perhaps this ought to have been expected. Removing the suspected archaizers from the training set makes the date model better, exactly as one would expect were they genuinely mislabeled. Nonetheless, the manifest's own labels are used as the primary specification throughout, since that variant is the most conservative and the only one requiring no judgment on my part.

Second, the two Atticizers displaced least, Strabo and Dionysius of Halicarnassus, are the earliest in the set, writing around the turn of the era when the Atticizing program was still young. The displacement grows with the maturity of the movement, which is what one would expect if it measures imitation rather than noise.

Additionally, text lengths remain uneven, from 19 passages to 540, so a short estimate is correspondingly noisy. The result is not driven by the short texts: weighting each text by its passage count gives -201 yr, and restricting to the 13 texts with at least 40 passages gives -206 yr.

What the Two Experiments Establish

The two experiments do not give the same number, and that difference is itself the substantive finding.

Total *lexical* archaizing in Hebrew buys 9 yr of apparent antiquity, while real *morphosyntactic* archaizing in Greek buys 206 yr. The gap is roughly an order of magnitude, and it falls where the feature-leverage analysis said it would. Swapping out diagnostic vocabulary moves a handful of the 250 weak lexical features and leaves the other half of the model's leverage untouched. The Atticizers did not swap vocabulary. They reconstructed a grammar, reviving the optative, restoring the $\mu\acute{\varepsilon}\nu \ldots \delta\acute{\varepsilon}$ articulation, writing $\tau\tau$ for $\sigma\sigma$, and matching Classical participial density. That is the coordinated multi-dimensional shift which the Hebrew sensitivity analysis identified as the only route to a displacement of centuries, and they managed it because an entire educational apparatus existed to teach them how.

The conclusion is therefore conditional rather than absolute, and it ought to be stated in that form. Archaizing can defeat a model of this class. What it takes is not a scribe who knows the old words, but one trained to write the old grammar and to sustain it across a whole composition. For any particular text, then, the question is whether such training existed in its milieu.

For the Hebrew Bible this becomes a historical question rather than a statistical one, and it is not settled here. The institutional preconditions do differ sharply between the two cases. The Second Sophistic had a documented curriculum of Attic prose composition, lexica of permitted usage, and a literary culture that rewarded successful imitation, whereas no comparable program of Classical Hebrew prose composition is attested for the Second Temple period. The Qumran evidence adduced by the dialectal school [5] is the closest analogue, and it is a good deal thinner. The sectarian scrolls mix CBH forms with Aramaisms rather than sustaining a consistent Classical grammar, which on the evidence here would displace their apparent date by tens of years rather than hundreds.

This bears directly on the archaic poems. An archaizing account of the early placement of the Song of the Sea or the Song of Deborah is not excluded by these results, but it carries a specific and demanding burden. It must posit sustained morphosyntactic imitation, of the kind that took the Second Sophistic a formal curriculum to achieve, rather than the lexical substitution that the CBH/LBH literature describes and that is measured above at 9 yr. No comparable institution is attested for Second Temple Judaism. The burden is therefore a heavy one, and it now has a number attached to it.

Discussion

Limitations

Interval width. The 68% intervals are ±137 yr and the 90% intervals ±281 yr. These are honest but wide, and they preclude any claim about position within the Persian period. Every substantive claim above is either a side-of-the-exile claim or an ordering claim.

Genre. Poetic texts are placed early and narrative texts late relative to their true dates. With one securely dated poetic book in the anchored corpus, genre cannot be separated from date by this design. This directly affects the archaic poems, which are the results most likely to be cited and are also the ones most exposed to this confound.

Corpus asymmetry. The anchored corpus is skewed post-exilic, because that is where external synchronisms are dense. Inverse-density weighting mitigates but does not remove the skew, and the pre-exilic end of the scale rests on 8 books. Still, extending the anchored corpus at the early end would do more for this model than any change of estimator.

The two archaizing measurements bound different things. Four substitution pairs is a small vocabulary of archaizing, so the Hebrew experiment prices only the lexical mechanism, and prices it at the extreme setting of total replacement. The Greek experiment prices real imitation but only at the intensity Second Sophistic authors actually achieved, in a language whose annotation is coarser than BHSA, and against a training set whose register classification is a scholarly judgment rather than a datum (S6 Table reports the sensitivity to that judgment). Neither experiment prices sustained morphosyntactic archaizing *in Hebrew*, which is the quantity a critic of the Song of the Sea result would most want. Simulating it would require a model of what Classical Hebrew grammar an imitator could reconstruct, and no such model exists.

The Greek transfer is an analogy rather than a measurement of Hebrew. That Greek imitators achieved a 206 yr displacement establishes that the deep mechanism works in principle and gives its rough scale. It does not establish that a Hebrew scribe could have done the same, and the claim that Second Temple Judaism lacked the institutional apparatus is a historical one, asserted here rather than demonstrated.

One database, one text tradition. All Hebrew results depend on the BHSA annotation of the Masoretic text. Consonantal variants in the Qumran and Septuagint traditions are not represented, and the Masoretic vocalization reflects a much later stage of the language than any composition date discussed here. The features used are drawn from the consonantal text and its morphological parse, but the parse itself is a modern scholarly product.

Scholarly Context for a Late Torah

The finding that the Pentateuchal sources are post-exilic is not novel in biblical scholarship, however contested it remains. A substantial tradition places the final composition of P in the Persian period [25, 26], and the "Persian period Pentateuch" position associated with the Copenhagen and Sheffield schools places the whole compositional process there [27, 28]. Against this, Hurvitz's linguistic argument for a pre-exilic P [1, 9] and the broader neo-documentary position maintain earlier dates.

What is offered here is not the direction of the result but its arrival with a calibrated uncertainty and a measured sensitivity to archaizing. The linguistic argument for a pre-exilic P has always had to assume that the post-exilic appearance of a text cannot be manufactured, while the argument for a post-exilic P has always had to assume the converse. The measurements above constrain both assumptions.

Extensibility

Given a morphologically annotated corpus and a set of externally anchored dates, the pipeline is indifferent to the language. The Greek replication was implemented from the same code with a different feature extractor. The binding constraint is annotation rather than method. Ugaritic, Phoenician, and epigraphic Hebrew would supply badly needed early anchors, but they lack comparable morphological parsing. Producing such a parse for the Iron Age epigraphic corpus would be the single most valuable contribution to this line of work.

Relation to Authorship Attribution

Attribution and dating are complementary. A method that assigns a text to a scribal school and a method that assigns it a date constrain different axes of the same question, and their errors are not the same errors. One natural extension would be to use attribution output as a register covariate in the date model, which would let the genre confound documented above be modeled rather than merely reported.

Conclusion

Biblical Hebrew morphosyntax carries a diachronic signal strong enough to order books in time and to tell pre-exilic from post-exilic composition, but too weak to resolve dates within the post-exilic period. Applied to the Pentateuchal sources, a model calibrated on independently anchored books places the Priestly, Deuteronomic, and JE composites after the exile with probability of at least 0.84, and recovers the classical sequence JE $\rightarrow$ D $\rightarrow$ P without having been trained on any of them. The Song of the Sea and the Song of Deborah are the earliest of the 19 undated units.

The premise on which all such conclusions rest, that a late author cannot convincingly write early, has been treated as an assumption for two decades. It is measurable, and measuring it shows that it is not one assumption but two, with different answers. Complete lexical archaizing of a securely dated late Hebrew book buys a mean of 9 yr of apparent antiquity, since the diagnostic vocabulary that the CBH/LBH literature treats as the substance of archaizing turns out to carry very little of a passage-level model's leverage. The Second Sophistic Atticizers, who reconstructed Classical grammar rather than Classical vocabulary and were trained by a formal curriculum to do so, are pushed 206 yr too early.

Archaizing therefore has a price, and that price is set by how deep the imitation reaches. This is a more useful result than immunity would have been, as it converts a rhetorical impasse into a comparison that can be made case by case. For any text whose date is disputed on linguistic grounds, the question is no longer whether archaizing is possible in the abstract, but whether the milieu in question supported the deep kind of it. For the Second Sophistic the answer is documented and affirmative. For Second Temple Judaism no comparable program is attested, and the closest analogue, the Qumran sectarian scrolls that mix Classical forms with Aramaisms rather than sustaining a Classical grammar, resembles the cheap mechanism rather than the expensive one.

Supporting Information

S1 Appendix. Source partitions. Chapter-and-verse specification of every documentary source, sub-stratum and poem dated in this study (Table 8).

S2 Table. Power analysis. Intraclass correlation, design effect and effective sample size by passage size (Table 9).

S3 Table. Configuration search. Combinations of passage size, feature set and calibration with out-of-sample metrics, including the uncalibrated variants (Table 10).

S4 Table. Feature leverage. Years of apparent date per standard-deviation shift for every feature, by family.

S5 Table. Greek corpus. All texts with dates, registers, genres and sources.

S6 Table. Register sensitivity of the Greek archaizing measurement. Displacement of the Atticizers under three definitions of the non-archaizing training set (Table 11).

Table 8. S1 Appendix. Source partitions. Chapter specification of every undated unit. The five Pentateuchal books as received are omitted (they are simply the whole book). Partitions follow the standard critical division; the three poems were additionally re-extracted at verse precision.

Unit	Chapters
P source	Genesis 1, 5, 6, 9, 11, 17, 23, 25, 35, 36, 46, 48; Exodus 6–7, 12, 16, 25–31, 35–40; Leviticus 1–27; Numbers 1–10, 13–20, 25–36
JE composite	Genesis 2–4, 6–8, 12–16, 18–22, 24–34, 37–45, 49–50; Exodus 1–5, 8–11, 13–14, 17–24, 32–34; Numbers 11–12, 21–24
D source	Deuteronomy 1–34
Genesis JE	Genesis 2–4, 6–8, 12–16, 18–22, 24–34, 37–45, 49–50
Exodus JE	Exodus 1–5, 8–11, 13–14, 17–24, 32–34
Numbers JE	Numbers 11–12, 21–24
D law code	Deuteronomy 12–26
D frame	Deuteronomy 1–11, 27–31, 33–34
Song of Moses	Deuteronomy 32
	Deut 32:1–43 at verse precision
Holiness Code	Leviticus 17–26
Leviticus P	Leviticus 1–16
Song of the Sea	Exodus 15
	Exod 15:1–18 at verse precision (see text)
Song of Deborah	Judges 5
	Judg 5:2–31 at verse precision
Jeremiah Dtr prose	Jeremiah 7, 11, 17–18, 21, 24–29, 32–45, 52

Table 9. S2 Table. Effective sample size by passage size. Passages within a book are correlated, so the raw passage count overstates the information available. $n_{\text{eff}} = n/(1 + (\bar{m} - 1)\rho_I)$ with ρ_I the median intraclass correlation across features.

Passage size	Passages	Median ICC	Design effect	n_{eff}	Gain vs $n = 25$
300	470	0.09	2.68	175.3	7.01x
500	285	0.13	2.38	119.6	4.79x
1000	145	0.20	1.96	74.0	2.96x

Table 10. S3 Table. Configuration search. Out-of-sample metrics for each combination of passage size, feature set and calibration under leave-one-book-out. "Coverage" is the empirical coverage of the nominal 68% interval; values near 1.0 indicate conservative intervals and values near 0.5 indicate the shrinkage problem discussed in the Model section.

Passage	Features	p	Calib.	MAE	ρ	Pair %	Coverage
1000	base	579	False	105	+0.56	69.5	0.60
1000	base	579	True	97	+0.56	68.5	0.64
1000	both	1268	False	94	+0.62	71.5	0.58
1000	both	1268	True	96	+0.62	71.9	0.68
1000	ngram	689	False	96	+0.59	70.8	0.53
1000	ngram	689	True	99	+0.60	70.8	0.67

Table 11. S6 Table. Register sensitivity of the Greek archaizing measurement. The displacement of the Atticizers depends on which texts are treated as non-archaizing training material. Variant A uses the corpus manifest's own labels; B additionally reclassifies authors independently described as classicizing in the literary-historical record; C discards register labels entirely and trains only on texts composed before 50 CE, prior to the Second Sophistic. Dates CE; a negative shift means the text is dated earlier than it was written.

Training set	Texts	Tested	MAE	ρ	Mean shift	Too early
A as-received manifest labels	42	14	145	+0.69	-206	11/14
B strict: classicizing authors reclassified	34	22	101	+0.80	-297	22/22
C date-blind: train only on ¡= 50 CE	26	11	120	+0.67	-397	11/11

Data Availability

All code, extracted feature matrices, and result files are available at the repository cited in the acknowledgments. The BHSA database is available under its own license from the Eep Talstra Centre for

Bible and Computer. Greek texts derive from the open First1KGreek and Perseus collections.

Acknowledgments

The author thanks the Eep Talstra Centre for Bible and Computer for the BHSA database and the Text-Fabric project for the access layer.

Competing Interests

The author declares no competing interests.

Funding

The author received no specific funding for this work.

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